

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Traversal Challenge

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	Assessment:	Assessment Tool 2 – Traversal Challenge
Weightage:	30% of CIA	Total Marks:	100
CO3 Statement:	Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, B-Trees, Red-Black Trees, and Splay Trees.		
Student Name:	P. Harsha vardhan	Reg. No.:	192411179
Section / Batch:	cse	Date:	31/07/20

Code of Conduct

I P. Harsha vardhan(192411179) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: harsha

Instructions

- This assessment tool contains 5 Traversal Challenge questions, Total: 100 Marks.
- Students can use any online/offline Traversal Challenge. Demonstrate tree.
- When a student answer using simulator, in evaluation the answer should have captured screenshots after every major operation
- Submit the report in PDF format with properly labelled screenshots as proof of completion.

Section / Topic	Focus	Q Nos	Marks
Tree Traversals	Traversal types, In order, Post order and Pre order	5	100
GRAND TOTAL:			100 Marks

Traversal Challenge

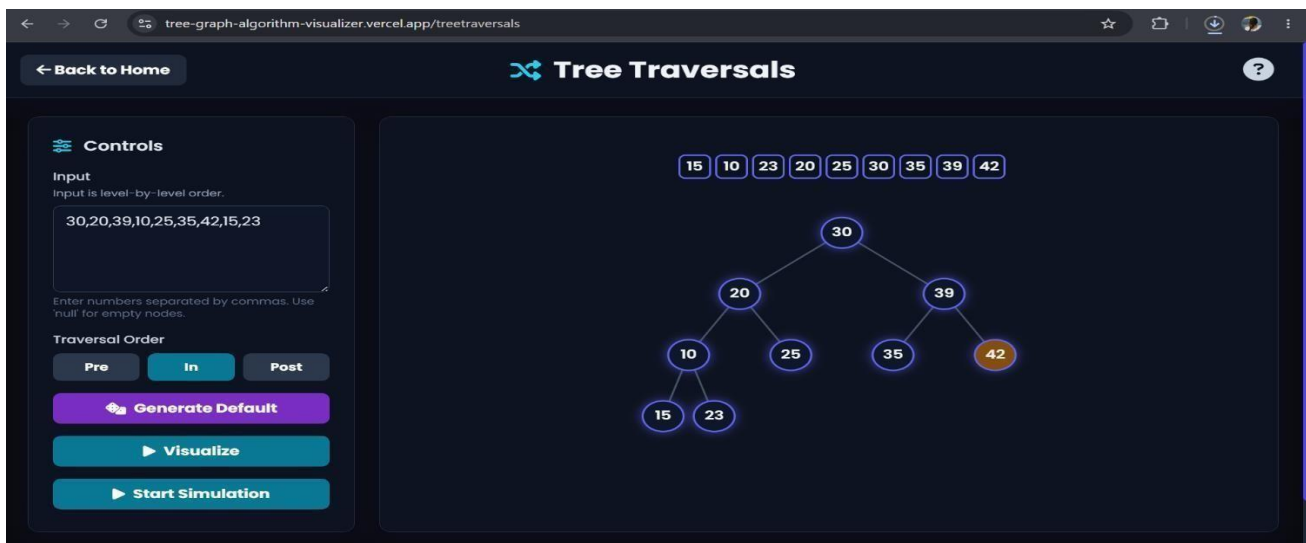
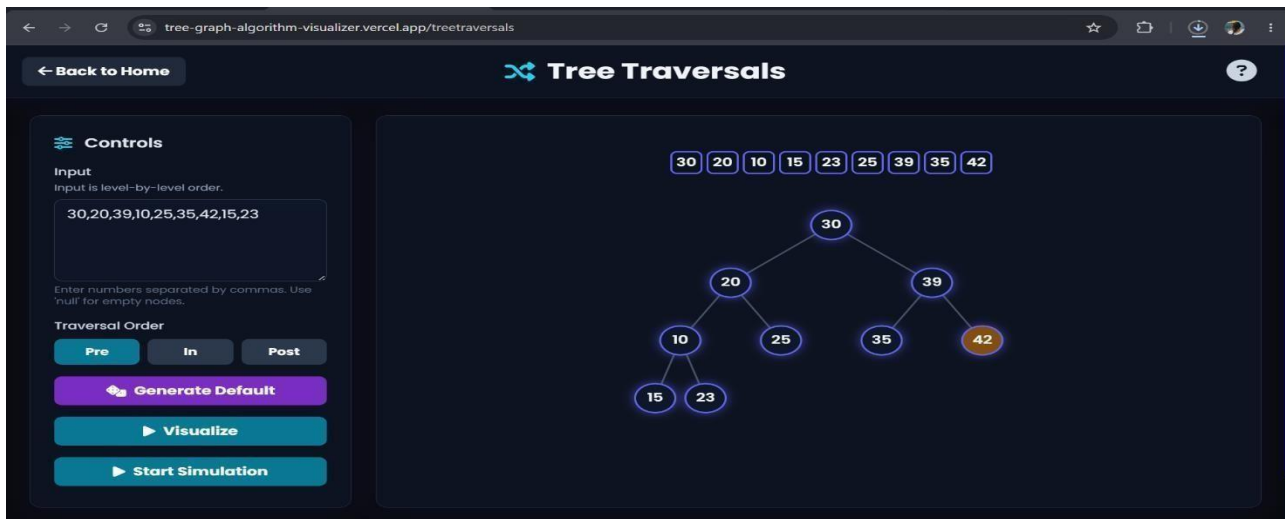
Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
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Conceptual Understanding	20	Demonstrates thorough understanding of all core Data Structures concepts; answers accurately reflect deep knowledge	Shows good understanding with minor conceptual gaps	Basic understanding; several misconceptions present	Limited understanding; major concepts misunderstood
Traversal Accuracy	20	Correctly completes all tree traversals (Preorder, Inorder, Postorder, Level Order) without errors.	Completes most traversals correctly with one minor error.	Correctly performs only some traversals with multiple errors.	Unable to perform most traversals correctly.
Selection of Appropriate Traversal	30	Correctly selects the appropriate traversal for every given problem scenario.	Selects the correct traversal for most scenarios.	Selects appropriate traversal for only a few scenarios.	Frequently selects incorrect traversal methods.
Completion & Time Efficiency	20	Completes all challenges within the allotted time while maintaining accuracy.	Completes most challenges within the time limit.	Completes only part of the challenge or exceeds the time limit slightly.	Unable to complete the challenge within the allotted time.
Evidence Submission	10	Uploads complete screenshots showing successful completion, score, and student details.	Uploads screenshots with minor missing information.	Uploads incomplete or unclear screenshots.	No valid evidence submitted.

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Question-1



Question-2

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Tree Traversals

1563247139181720

```
graph TD; 15((15)) --- 6((6)); 15 --- 18((18)); 6 --- 3((3)); 6 --- 7((7)); 3 --- 2((2)); 3 --- 4((4)); 7 --- 13((13)); 7 --- 9((9)); 18 --- 17((17)); 18 --- 20((20)); style 20 fill:#f96
```

Input
Input is level-by-level order.

15,6,18,3,7,17,20,2,4,13,9

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre In Post

Generate Default

Visualize

Start Simulation

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Tree Traversals

2346137915171820

```
graph TD; 15((15)) --- 6((6)); 15 --- 18((18)); 6 --- 3((3)); 6 --- 7((7)); 3 --- 2((2)); 3 --- 4((4)); 7 --- 13((13)); 7 --- 9((9)); 18 --- 17((17)); 18 --- 20((20)); style 20 fill:#f96
```

Input
Input is level-by-level order.

15,6,18,3,7,17,20,2,4,13,9

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre **In** Post

Generate Default

Visualize

Start Simulation

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Tree Traversals

2431397617201815

```
graph TD; 15((15)) --- 6((6)); 15 --- 18((18)); 6 --- 3((3)); 6 --- 7((7)); 3 --- 2((2)); 3 --- 4((4)); 7 --- 13((13)); 7 --- 9((9)); 18 --- 17((17)); 18 --- 20((20)); style 15 fill:#f96
```

Input
Input is level-by-level order.

15,6,18,3,7,17,20,2,4,13,9

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre In **Post**

Generate Default

Visualize

Start Simulation

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Question-3

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Tree Traversals

?

Input

Input is level-by-level order.

1,2,3,4,5,6,7

Enter numbers separated by commas. Use 'null' for empty nodes.

Pre

In

Post

Generate Default

Visualize

Start Simulation

4 2 5 1 6 3 7

1

2

3

4

5

6

7

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Tree Traversals

?

Input

Input is level-by-level order.

1,2,3,4,5,6,7

Enter numbers separated by commas. Use 'null' for empty nodes.

Pre

In

Post

Generate Default

Visualize

Start Simulation

1 2 4 5 3 6 7

1

2

3

4

5

6

7

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Tree Traversals

?

Input

Input is level-by-level order.

1,2,3,4,5,6,7

Enter numbers separated by commas. Use 'null' for empty nodes.

Pre

In

Post

Generate Default

Visualize

Start Simulation

4 5 2 6 7 3 1

1

2

3

4

5

6

7

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QUESTION-4

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Tree Traversals

?

Input

Input is level-by-level order.

7,6,5,4,3,2,1

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre In Post

Generate Default

Visualize

Start Simulation

4 3 6 2 1 5 7

```
graph TD; 7((7)) --- 6((6)); 7 --- 5((5)); 6 --- 4((4)); 6 --- 3((3)); 5 --- 2((2)); 5 --- 1((1));
```

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Tree Traversals

?

Input

Input is level-by-level order.

7,6,5,4,3,2,1

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre In Post

Generate Default

Visualize

Start Simulation

7 6 4 3 5 2 1

```
graph TD; 7((7)) --- 6((6)); 7 --- 5((5)); 6 --- 4((4)); 6 --- 3((3)); 5 --- 2((2)); 5 --- 1((1));
```

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Tree Traversals

?

Input

Input is level-by-level order.

7,6,5,4,3,2,1

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre In Post

Generate Default

Visualize

Start Simulation

4 6 3 7 2 5 1

```
graph TD; 7((7)) --- 6((6)); 7 --- 5((5)); 6 --- 4((4)); 6 --- 3((3)); 5 --- 2((2)); 5 --- 1((1));
```

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Question-5

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Tree Traversals

?

Input

Input is level-by-level order.

7,5,1,8,3,6,0,9,4,2

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

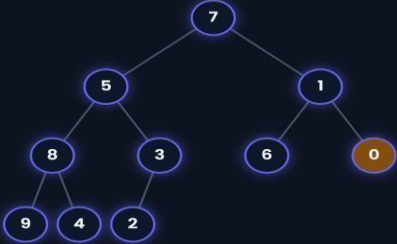
Post

Generate Default

Visualize

Start Simulation

7 5 8 9 4 3 2 1 6 0



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Tree Traversals

?

Input

Input is level-by-level order.

7,5,1,8,3,6,0,9,4,2

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

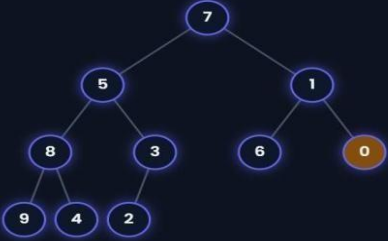
Post

Generate Default

Visualize

Start Simulation

9 8 4 5 2 3 7 6 1 0



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Tree Traversals

?

Input

Input is level-by-level order.

7,5,1,8,3,6,0,9,4,2

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

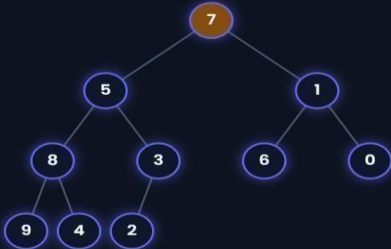
Post

Generate Default

Visualize

Start Simulation

9 4 8 2 3 5 6 0 1 7



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